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EDITED BY

Maomao Zhang,
Huazhong University of Science and
Technology, China

REVIEWED BY

Shaohong Wu,
Chinese Academy of Sciences
(CAS), China
Xiaoxuan Wu,
China University of Geosciences
Wuhan, China

*CORRESPONDENCE

Xin Han,
✉ 2023101440@niit.edu.cn

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Interactive loops between anthropogenic urban-rural construction and natural systems in China: quantifying green development pathways and policy effectiveness

Xin Han^{1,2,3*}

¹Institute of Silk Road Economics and Management, Xinjiang University of Finance and Economics, Ürümqi, Xinjiang, China, ²School of Business and Trade, Nanjing University of Industry Technology, Nanjing, Jiangsu, China, ³Jiangsu Rongzer Information Technology Corp, Nanjing, Jiangsu, China

Introduction: Accelerating urbanization and rural construction (URC) in China profoundly alter natural systems (land cover, ecosystems, climate regulation). The existing green development index (GDI) systems remain inadequate in capturing anthropogenic impacts on natural ecosystems and lack an integrated framework that combines remote sensing and statistical data. Using these combined parameters, this study examines bidirectional interactions between anthropogenic construction activities and natural processes.

Methods: We calibrate the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS)-Entropy Weight calculation to measure the value of 286 cities or higher than the prefecture level from 2010 to 2023. The GD level of URC in China in 2023 is further by the clustering method. Finally, the deep Q-network (DQN) model is developed to simulate the policy responses under two scenarios of nature-society interactions.

Results: Results show that the standardized GDI of URC increases from 0.404 in 2010 to 0.553 in 2023. Four clusters are identified, including Advantaged (29 cities), Second best (68 cities), Average (120 cities), and Weak (69 cities). DQN simulations indicate that increasing renewable energy by 20% improves GDI more than a 5% reduction in emissions. The renewable energy pathway is preferred in over 94% of episodes across sensitivity tests.

Discussion: The study recommends scaling renewable energy integration in Average and Weak clusters, while Advantaged cities play roles in carbon-neutral practices.

KEYWORDS

China, deep Q-network, entropy, green development index, TOPSIS, urban and rural

1 Introduction

Human interactions on urbanization and rural reconstruction in large economies like China profoundly alter natural processes through land use expansion, resource consumption, and emissions. This imposes significant constraints on socio-economic development, public health, and long-term resilience (Leng et al., 2024; Kai et al., 2024). Understanding these bidirectional interaction mechanisms is crucial for achieving sustainable development, as

anthropogenic pressures on natural systems trigger environmental responses that, in turn, reshape societal pathways.

Green development (GD) is the common theme of human development in the modern era and the key path of China's economic transformation (Huang et al., 2024). Policymakers, international organizations, and researchers prioritize ensuring economic growth without increasing carbon emissions, among other climate concerns (Dinh-Quoc et al., 2025). Previous literature highlighted the critical necessity for China, the globe's second-largest nation and leading emissions contributor, to seek sustainable growth (Ibrahim et al., 2022). Therefore, GD is the inevitable choice and fundamental way out for China (Fang et al., 2012). Urban and rural development is an important vehicle for promoting GD and building a beautiful China. In October 2021, the General Offices of the CPC Central Committee and the State Council released Opinions on Promoting the GD of URC. In compliance with the principles, local administrations at all levels have implemented a variety of strategies to support urban and rural GD. Several studies have also begun to study the GD of URC. Liu et al. (2025) examined the global-scale expansion of urban–rural settlements at fine spatial resolution from 1985 to 2020. A comprehensive framework of farmland fragmentation indices was constructed using a geographically weighted regression model (Zhou et al., 2022). The model's explanatory power at the village and commune levels remains limited. In addition, due to data constraints, this study only includes a total of 26 cities in the Yangtze River Delta. Therefore, the proposed index system cannot yet be readily applied to planning at larger spatial scales. Based on data from the main urban area of Hangzhou, Li S. et al. (2024) investigated how different urban and rural construction (URC) land expansion processes affect varying levels of urban–rural disparities. However, this study does not incorporate factors related to residents' needs and green lifestyles. The use of statistical crop yield data limits the ability to capture urban–rural differences in settlement-induced yield losses. Limitations in the temporal coverage and spatial resolution of available data are also common constraints in previous studies. Addressing these limitations could help facilitate coordinated economic and environmental development, improve agricultural production, and promote the formation of new urban development models. A spatial distance clustering method was deployed to divide key URC areas into statistical units (Jiang et al., 2024). However, the quantification of the spatial impacts of URC was constrained by inconsistencies in the sizes of these key areas.

To our knowledge, current research on this topic uses different index sets with fewer criteria, or studies at different levels (provincial level) that do not quantify GD in rural and urban construction at the city level or by cluster. Also, another significant research gap in existing studies is the limited integration of statistical data with remote sensing data. Remote sensing enables objective quantification of environmental responses to human activities. This allows tracking built-up area expansion, forest coverage changes, cultivated land loss, and green space distribution, key indicators in GDI establishment. To address this research gap and more effectively examine the progress of GD in URC in China, this paper aims to establish a comprehensive set of index systems. These systems will quantify the GD of URC across various cities using a unified standard and will be used to analyze the distinctive features of the GD of URC in China. By integrating remote sensing with

statistical data, the GDI system becomes more comprehensive, enabling policymakers to better regulate URC. This research is critical to confronting ecological problems amid China's rapid economic growth, as well as ensuring long-term socioeconomic stability.

Therefore, the construction of GDI in URC should fully absorb relevant research results and make necessary adjustments. On the one hand, the carrier of this study is urban data. The GDI of URC should take into account the commonness of each city. For the unavailable variable data in the GDI system, and the inadaptability indicators, most of them should be selected. Furthermore, according to the general requirements of the opinions, the framework of the GDI system should adhere to the priority of ecology, conservation, and protection, the system's principles, sustainable growth and security, and simultaneously foster the construction of material and ecological civilization. Also, the system should implement the target of carbon peak and carbon neutrality, promote the action of renewal URC, and accelerate the transformation of the mode of URC. This study fills a research gap by supporting overall green evolution in social and economic development, and by laying the groundwork for thoroughly establishing a modern socialist country. The creation of GDIs for URC should continue to focus on resource use, governance of the environment, quality of ecology, preservation of the environment, development effectiveness, and green life from various viewpoints.

The remaining sections are outlined as follows. In Section 2, we provide an overview of pertinent works concerning the GD of URC, including relevant criteria sets and implemented methodologies. In Section 3, the focus shifts to data selection, index set selection, quantification methods, data dimensionality reduction, analysis, clustering techniques, and the DQN. Section 4 will detail the analysis outcomes and offer policy recommendations. Lastly, the study will conclude by summarizing the findings.

2 Literature review

2.1 Conceptual foundations

GD is traditionally defined as a process of development that integrates high economic growth, increased social equity, and sustained environmental protection using fewer resources and causing less environmental deterioration (Kousar et al., 2023; Kai et al., 2024; Leng et al., 2024). In China, GD is framed around the familiar principle of meeting today's needs without shortchanging future generations, with a strong emphasis on ecological civilization, carbon neutrality targets, and rebuilding a healthier relationship between people and nature (Beekman, 2004).

When applied to urban–rural construction (URC), this idea moves into the built environment. Drawing from the General Offices of the CPC Central Committee and State Council's guiding opinions, GD of URC refers to a broad transformation of how construction activities are planned and carried out, spanning land use, infrastructure, buildings, and public services, to make them more resource-efficient, low-carbon, ecologically sound, and socially inclusive. Rather than treating urban expansion and rural development as separate tracks, URC-GD sees them, along with their interactions (Ma and Shi, 2023) with natural systems like

land cover, ecosystems, and climate regulation, as parts of a single, interconnected socio-ecological process (Kuang et al., 2018).

The importance of assessing the URC within an integrated indexing system is based on the bidirectional feedback loops between human processes and natural systems (Xiao et al., 2017). The urban-rural integration theory argues that disconnected development increases spatial imbalance, friction of resource use, and ecological damage. A combined indicators system is needed to capture both the positive benefits (green infrastructure) and the negative trade-offs between the expansion of built-up areas *versus* the loss of cultivated land across different scales (Zhang, 2025). This aligns with China's environmental protection strategy and the UN Sustainable Development Goals (SDGs), particularly SDG 11 (sustainable cities and communities) and SDG 15 (life on land).

The external impacts of URC activities, including air pollution, noise, and ecological degradation, need policy intervention (Huang et al., 2024).

2.2 Existing studies on GD of URC

GD in URC aims to capture the contradictions associated with economic development by minimizing the consumption of resources, while simultaneously improving environmental and ecological governance, thereby operationalizing the principle of sustainable development (Clausen and Rudolph, 2020). Some recent studies have extensively discussed this issue. The first direction focused on specific regional studies, such as the Urban GDI of the Yangtze River Economic Belt (Pan et al., 2024) and the rural GD in the Yellow River Basin (Chen et al., 2020). Secondly, subjective and objective research methods complement each other. Subjective research employs methods like weighting, expert ratings, and ESG evaluation systems (Rahdari, 2016), and the development of GDIs (Musa et al., 2019). Although subjective methods are subject to human influence, they possess scientific value. To ensure objectivity, studies often used econometric models, including principal component analysis (PCA), the Entropy weight method, and cluster analysis (Liu et al., 2021; Shemshadi et al., 2011). Several studies used multi-dimensional measurement models like the combination weighting method based on entropy and grey correlation coefficients (Song et al., 2024). Thirdly, the development of the GDI systems concentrated on six key domains encompassing resource utilization, environmental regulation, air quality, ecological preservation, economic effectiveness, and green living. This was influenced by the GDI System formulated jointly by the Chinese Bureau of Statistics and the Ministry of Environmental Protection in December 2016. Previous studies have mainly built on these six aspects in their research on GDI systems (Xu et al., 2022; Yang et al., 2019).

A typical study systematically reviewed the evolving characteristics of GD levels across China's provinces between 2006 and 2017, considering the permitted national GDI with certain modifications (Xu et al., 2022). The findings indicated that GD levels in China's provinces had been increasing. The core concept of GD and a hierarchical evaluation approach, combined with the entropy method, were examined to create an urban GD assessment index system. This system includes five key dimensions: living environment improvement, pollutants management, ecological efficiency, economic growth optimization, and innovative potential

(Wang et al., 2018). This measure framework was then deployed to assess the GD levels of nine cities in a river delta region. A framework was established for evaluating enterprise GD performance by merging the concepts of GD with enterprise performance measurement (He et al., 2021). Using tabular array data from 458 Chinese corporations between 2011 and 2018, the study employed functional data analysis and the functional entropy weight approach to mimic the GD index of these enterprises. Xiao et al. (2022) introduced a novel approach by utilizing a combined entropy weighting approach to assess the GD hierarchy in Hubei Province between 2010 and 2018. Subsequently, they deployed the Durbin model to delve into the temporal and spatial dynamics and the diffusion effects. The findings revealed a significant increase in GD levels post-2014, accompanied by a gradual reduction in regional disparities. An enhanced evaluation framework was proposed to address existing gaps and to advance understanding by examining the status, imperfections, and spatiotemporal disparities of Industrial GD (IGD) (Li et al., 2020). Using 30 province-based panel data between 2006 and 2015, the study illustrated several key findings that emerged from the implemented framework: both the performance and standard of IGD experienced enhancements, yet the growth rate diminished considerably in the ensuing years.

The TOPSIS-Entropy model, along with the Theil inequality, was calibrated to assess disparities in industrial GD across three major metropolitan areas (Wang et al., 2021). Additionally, component and vertical clustering analyses were utilized to examine and juxtapose the business ecological efficiency within these metropolitan areas. The research revealed that the Pearl River Delta demonstrated the most advanced level of industrial GD, whereas the Yangtze River Delta exhibited the lowest level. Notably, Beijing, Shanghai, and Shenzhen stood out with the highest industrial GD levels among the studied cities. Another study focused on calibrating and optimizing the parameters of the traditional Water Resources Ecological Footprint (WEF) model together with a robust WEF model tailored for diverse regions (Yang et al., 2023). Additionally, a hybrid WEBR model was introduced to illustrate the correlation between water resource sustainability and socioeconomic factors. The combined model incorporating the above techniques was applied and validated in the Beijing-Tianjin-Hebei (BTH) rectangular.

An innovative concept of development was used as a basis to construct a holistic assessment indicator system involving six components: development interactions, industrial and life systems, ecological, carbon-neutral, cyclical, and economic benefits (Lu et al., 2024). The game theory integrated robust VIKOR model was employed for comprehensively assessing the construction degree of the clean, carbon-neutral, and loop development monitoring system using panel data from some provinces in the southwestern region between 2012 and 2021. To examine the environmental economic transformation in Chinese provinces since the reformatory period, a dynamic indicator was constructed using the slacks-based measure (SBM) mechanism (Chen, 2015). The investigation revealed variability and inconsistency in each province's evaluation, implying that China's ecological economic transformation remained in an unstable early stage. It highlighted the importance of enacting more appropriate ecological governance to encourage China's long-term sustainable growth.

2.3 Trade-offs between scale indicators and methodological approaches

A fundamental challenge in GD research involves balancing the scale of research with the selection of indicators. Regional or provincial-level studies, including the Yangtze River Economic Belt (Luo et al., 2022; Pan et al., 2024), the Yellow River Basin (Niu and Mao, 2025), or the Pearl River Delta (Wang C. et al., 2024), often allow the incorporation of highly context-specific and granular indicators, such as mountain-city landscape fragmentation (Wang et al., 2023), local farmland fragmentation patterns (Das et al., 2025), or fine-scale urban–rural land-use gradients (Li et al., 2025). Specific circumstances, such as geography and climate, can impact the way that socio-economics regionalizes the effect of GD pathways in regard to studying localized approaches. However, these methods have limited compatibility and are not helpful for formulating nationally coordinated strategies.

The research conducted on a national level may also analyze and compare the data more comprehensively. The research may use standardized metrics based on data from statistical yearbooks and remote sensing data. Though this method provides a comprehensive evaluation of data for hundreds of cities, it is often accompanied by a lack of sensitivity to regional differences. For instance, the research may not take into account regional environmental challenges such as ecological vulnerability in the west of China due to high elevations or industrial pollution in the east. The research may also not take into account regional disparities in land use efficiency and ecosystem services between urban and rural areas.

Another critical trade-off concerns data sources and integration. The majority of the research conducted previously only relies on socio-economic statistical data. The data provides good temporal resolution but is accompanied by a high level of administrative aggregation bias and is not updated in a timely fashion (Chen et al., 2017; Piras and Pedes, 2025; Chen et al., 2025). On the other hand, remote sensing-based indicators such as forest coverage, PM_{2.5} concentration, and built-up area expansion based on Landsat, Gaofen, and GlobeLand30 data provide an objective assessment of the influence of human activities on natural systems. However, the integration of these two data types is limited in previous research due to the different resolutions and standards of the data. Therefore, the majority of the GDI methods underestimate the complex interaction between construction activities and natural systems such as cultivated land conversion, green space fragmentation, and road area expansion.

From a methodological perspective, traditional single-method approaches (single entropy weight or basic TOPSIS) often struggle to balance multi-dimensional criteria while maintaining objectivity (Zou et al., 2023; Wang L. et al., 2024). Studies focusing only on measurement rarely extend to dynamic policy simulation, leaving a gap in understanding long-term policy effectiveness under different nature–society interaction scenarios (Wu, 2025; Zellner, 2024). Additionally, the selection of indicators frequently faces a universality–specificity dilemma: overly general indicators enhance comparability but reduce explanatory power for heterogeneous Chinese cities; overly specific indicators improve depth but compromise the ability to conduct nationwide benchmarking.

2.4 Contributions of this study

This study addresses these gaps and makes the following distinct contributions to the literature:

First, it constructs a comprehensive and unified GDI system for URC that integrates six first-order indicators and 42 second-order indicators, drawing from the national GDI System while incorporating remote-sensing-derived metrics (e.g., forest coverage from GlobeLand30, PM_{2.5} from MODIS/MERRA-2, built-up area and road-area increments from Landsat and Gaofen series). This deep data fusion enhances the objectivity and spatial precision of environmental quality and ecological protection indicators, overcoming the limitations of purely statistical approaches.

Second, by applying the TOPSIS-Entropy Weight Method to evaluate 286 prefecture-level and above cities over the period 2010–2023, this study provides one of the most extensive national-scale assessments to date. It not only reveals the overall upward trend in URC-GD but also identifies environmental quality as the dominant influencing factor and classifies cities into four distinct clusters (Advantaged, Second-best, Average, and Weaker) using PCA and Ward's clustering method. This classification provides an advanced understanding of spatial heterogeneity beyond the provincial aggregates used in previous studies.

Third, the study innovates in terms of methodology through the application of the TOPSIS Entropy approach in conjunction with the DQN reinforcement learning model. Unlike previous static measurement studies, the DQN simulates policy responses under two interactive scenarios—increasing renewable energy adoption by 20% *versus* reducing emissions by 5%—and demonstrates that renewable energy policies yield superior effectiveness across different city types. This dynamic simulation component represents a clear marginal contribution, as it bridges the gap between evaluation and policy optimization, directly supporting the implementation of the 2021 Opinions on Promoting the GD of URC.

3 Data description and methodologies

3.1 Data acquisitions and indexing

Urbanization is an unavoidable aspect of modernization, with considerable potential demand in China. It is critical for sustaining a prosperous and secure socioeconomic development, establishing inventive growth sequences, and enhancing shared wealth. The 18th National Congress has publicly launched an innovative urbanization program, emphasizing a more compassionate, ecological, and intellectual strategy for urban development (Huang et al., 2024). The data was curated from multiple sources, including National Bureau of Statistics of China (2024), NASA (2024), Leng et al. (2024), and Yang and Tang (2024). Cities above the prefecture level in China between 2010 and 2023 are selected for analysis. Fine particulate matter (PM_{2.5}) data are obtained from MODIS combined with MERRA-2 reanalysis, enabling consistent spatiotemporal coverage across 286 cities. Land use/land cover metrics such as forest coverage, green space in built-up areas, built-up area increment, road area increment, and cultivated land change were extracted from satellite databases, including Landsat, Gaofen

series, GlobeLand30. These parameters were converted from remote sensing data into tabular form using Python and subsequently used for TOPSIS calculations.

Specifically, we utilize six common first-order indicators and a total of 42 second-level indicators as indicated in Table 1. The distribution of secondary indexes is resource utilization (10 items), environmental governance (6 items), environmental quality (7 items), ecological protection (8 items), growth quality (5 items), and green life (6 items).

3.2 Research methodologies

3.2.1 TOPSIS-entropy weight method

The extreme value method is an effective approach for establishing sustainable development indicators (Xu et al., 2020; Vu-Thi-Minh et al., 2026). Extreme value and entropy weight techniques are employed to determine the weights of the target and each criterion layer. We implement the following procedures. Firstly, we construct the original data matrix and utilize extreme values to normalize the data. Secondly, we design a TOPSIS-Entropy Weight Method to identify criterion weights with greater independence. This method seeks to address the subjectivity inherent in allocating criterion weights in the typical TOPSIS approach. Finally, we employed the TOPSIS-entropy weight method to determine the weights of 42 specific indicators in the evaluation index set for urban and rural GDP. Below are detailed procedures for calibrating the TOPSIS-Entropy weight method.

Step 1: Construct the original matrix (see Equation 1)

$$\Phi = \begin{pmatrix} V_{11} & \cdots & V_{1n} \\ \vdots & \ddots & \vdots \\ V_{m1} & \cdots & V_{mn} \end{pmatrix} \quad (1)$$

Wherein, Φ represents all variables involved in calculating the GDI of URC, V_{mn} represents the value of V variable in the year n of the city m .

Step 2: Extreme value standardization

According to the contents listed in Table 1, it can be seen that among the 42 indicators, there are 6 indicators with different units and 10 indicators with different directions. Therefore, the dimensions and orders of magnitude of each index of the green evaluation system for URC are not uniform. In order to eliminate the impact brought by the dimensional differences and orders of magnitude differences, it is essential to homogenize the heterogeneous indexes and standardize them, so as to solve the homogeneity of various heterogeneous indexes. Meanwhile, indicators with positive and negative connotations convey different implications: higher values of positive or lower values of negative connotations indicate better performance. Thus, data normalization is achieved using the extreme value method, as illustrated below.

Positive indicators (see Equation 2):

$$V'_{mn} = \frac{V_{mn} - V_{\min}}{V_{\max} - V_{\min}} \quad (2)$$

Negative indicators (see Equation 3):

$$V'_{mn} = \frac{V_{\max} - V_{mn}}{V_{\max} - V_{\min}} \quad (3)$$

where V'_{mn} represents the value of V variable after the standardization of the extreme value of the n index in the m city, and $V_{\max}$ denotes the maximum value of V variable. $V_{\min}$ is the minimum of V . In order to eliminate the number 0 generated by the above standardization, it is necessary to shift the coordinates to the left by 0.001 units to get a new one (see Equation 4).

$$V'_{mn} = V'_{mn} + 0.001 \quad (4)$$

Step 3: Calculate the proportion of each variable V of city m , which is denoted as Equation 5.

$$P_{mn} = \frac{V'_{mn}}{\sum_{m=1}^m V'_{mn}} \quad (5)$$

Step 4: Formulate the entropy value of the n index as Equation 6:

$$E_n = -\frac{1}{\ln(286)} \sum_{m=1}^m (P_{mn} * \ln(P_{mn})) \quad (6)$$

Step 5: Formulate the difference coefficient for the n indicator as Equation 7:

$$C_n = 1 - E_n \quad (7)$$

The lower the entropy value, the higher C_n , and the higher the impact on the evaluation results.

Step 6: Calculate the weight value with the following formula (see Equation 8):

$$W_n = \frac{C_n}{\sum_{n=1}^n C_n} \quad (8)$$

The weights of each first-level index of the GDI are, respectively, resource utilization (16.3093%), environmental governance (10.687%), environmental quality (13.0872%), ecological protection (46.621%), growth quality (5.43332%), and green life (7.86218%).

Step 7: Integrate index weights x_{ij} and x_{ij} into the aggregated weight. The resulting weighted decision matrix is calculated using Equation 9

$$Y = (y_{ij})_{m \times n} \quad (9)$$

Step 8: Deploy extreme values of each index to form the optimal scheme Y^+ and the worst scheme Y^- (see Equations 10, 11), respectively.

TABLE 1 Index system of urban and rural GDI.

First-order index	Secondary index	Unit	Direction	Weight (%)
Resource utilization	Incremental power usage per GDP unit	kW-hours/10 ³ CNY	Negative	0.153
	Carbon dioxide per GDP unit	Ton/10 ⁴ CNY	Negative	0.091
	Household water usage ratio	%	Positive	0.124
	Water consumption per GDP unit	m ³ /10 ³ CNY	Negative	0.131
	Industrial waste pollutants per GDP unit	Ton/10 ⁴ CNY	Negative	0.491
	The proportion of water used for the production and operation of the total water supply	%	Negative	0.089
	Increase in urban construction land area (1 in 2005)	%	Positive	2.388
	Utilization rate of reclaimed water	%	Positive	0.959
	Gas penetration rate	%	Positive	5.202
	Water supply penetration rate	%	Positive	6.682
Environmental governance	The density of pipelines for drainage (1 in 2005)	%	Positive	1.462
	Family garbage's harmless remediation rate	%	Positive	0.172
	The sewage processing factory's centralized rate	%	Positive	0.807
	Garbage removal efficiency	Tons/car	Positive	3.709
	Mechanization rate of road cleaning	%	Positive	3.046
	The proportion of public toilets above the third-class	%	Positive	1.491
Environmental quality	Rate of change of the average concentration of fine particulate matter	%	Negative	0.067
	Rate of change of industrial sulfur dioxide emissions	%	Negative	0.059
	Rate of change of industrial wastewater discharge	%	Negative	0.071
	Change rate of industrial soot emission	%	Negative	1.512
	Fertilizer applied per unit of cultivated land	Ton/ha	Positive	4.577
	The proportion of water used for public services in the total water supply	%	Positive	6.727
	Sewage discharge increment (2005 fixed base)	%	Negative	0.074
Ecological protection	Forest coverage	%	Positive	7.428
	Green space coverage in built-up areas	%	Positive	0.334
	Road area increment (2005 is the fixed base)	%	Positive	31.726
	Nature reserve area proportion	%	Positive	0.586
	Proportion of groundwater in the public water supply	%	Positive	1.436
	Change rate of the mean value of cultivated land	%	Positive	1.361
	The percentage of soil devoted to amenities for the public to the overall surface of urban construction land	%	Positive	1.923
	Built-up area increment (based on 2005)	%	Positive	1.828
Quality of growth	Per capita GDP growth rate	%	Positive	0.079
	Urbanization rate	%	Positive	3.813
	The growth rate of savings of urban and rural residents	%	Positive	0.420
	The percentage of the tertiary sector's value-added to GDP	%	Positive	0.4096
	Secondary industry's percentage of employees	%	Positive	0.712

(Continued)

TABLE 1 Continued

First-order index	Secondary index	Unit	Direction	Weight (%)
Green life	Install street lighting ratio	%	Positive	0.847
	Increase in public library books <i>per capita</i> (2005 as the base)	%	Positive	1.969
	Increase in hospital beds per 10,000 people (2005, fixed basis)	%	Positive	2.556
	Increase in the number of employees in public administration and social organizations (2005 base)	%	Positive	0.703
	Proportion of pavement area	%	Positive	0.702
	The proportion of parks' green areas	%	Positive	1.086

$$Y^+ = (y_1^+, y_2^+, \dots, y_n^+) \quad (10)$$

$$Y^- = (y_1^-, y_2^-, \dots, y_n^-) \quad (11)$$

where $y_1^+ = \max\{y_{1j}, y_{2j}, \dots, y_{mj}\}$, $y_1^- = \min\{y_{1j}, y_{2j}, \dots, y_{mj}\}$, $j = 1, 2, \dots, n$

Step 9: Compute the total distances between each scheme as follows (Equations 13, 14).

$$D_i^+ = \left[\sum_{j=1}^n (y_{ij} - y_j^+)^2 \right]^{1/2} \quad (13)$$

$$D_i^- = \left[\sum_{j=1}^n (y_{ij} - y_j^-)^2 \right]^{1/2} \quad (14)$$

Step 10: Compute the relative proximity of each scheme (see Equation 15)

$$S_i = D_i^- / (D_i^+ + D_i^-) \quad (15)$$

$$i = 1, 2, \dots, m$$

Step 11: Calculate the GDI of URC as Equation 16

$$URCGDI_{mn} = \sum_{n=1}^n W_n * P_{mn} \quad (16)$$

3.2.2 Deep Q-network (DQN) procedures

The DQN model optimizes GDI policies by selecting from a discrete action space of two actions: increasing renewable energy by 20% and reducing emissions by 5%. The state space comprises GDI and the six indicators (RU, E.G., EQ, EP, QG, GL), capturing each city's environmental and socio-economic profile. The reward function is defined as $\text{Reward} = \Delta\text{GDI} + \alpha \cdot \Delta\text{EQ} - \beta \cdot \text{Cost}$, where ΔGDI is the GDI increase, ΔEQ is the change in Environmental Quality, and Cost represents implementation expenses. This prioritizes EQ due to its significant weighting in the GDI framework.

The DQN employs a neural network with three fully connected layers: an input layer with 8 state variables (GDI and six indicators),

two hidden layers with 64 and 32 nodes, and an output layer with 2 action values. The ReLU activation function is used for the hidden layers due to its efficiency and ability to mitigate the vanishing gradient problem (Ta et al., 2025a; Bui-Minh et al., 2025; Le, 2025). The model is trained over 1000 episodes, each with 50-time steps, to simulate long-term policy impacts. Training utilizes experience replay with a 1,000-transition buffer, a discount factor (γ) of 0.99, and an epsilon-greedy policy (initial $\epsilon = 1.0$, decaying to 0.01) for exploration. The target network is updated every 100 steps to stabilize learning, and the Adam optimizer is applied with a learning rate of 0.001. The DQN estimates Q-values to select the action with the highest expected reward for a given state, with policy impacts modeled via a simulation function integrated into the dashboard's policy simulation plot (Le and Xuan-Thi-Thu, 2024).

The TOPSIS-Entropy Weight Method provides essential technical foundations for the DQN's design and execution, specifically supporting the state space definition, training, reward function formulation, and policy simulation. By providing a weighted set of indicators, it ensures that the input vector accurately represents the relative importance of each indicator. The high weight of EQ guides the DQN to prioritize environmental quality in its state representation, enabling the neural network to focus on critical sustainability factors during training and action selection. Also, by highlighting EQ's significant weight, the method navigates the inclusion of ΔEQ in the reward function. Thus, it can support the DQN prioritize policies with substantial environmental quality benefits. Finally, by categorizing cities based on their GDI performance, the method enables the DQN to tailor actions to specific urban profiles. For example, high-GDI cities can be prioritized for renewable energy policies, while low-GDI cities may benefit more from emission reductions. This enhances the efficiency of the epsilon-greedy exploration strategy, allowing the DQN to focus on relevant actions during training and produce suitable policy recommendations.

4 Results and discussion

4.1 The overall analysis of the national URC GD

Using the TOPSIS-Entropy Weight calculation, the relevant formula of the GDI of URC can be calculated respectively

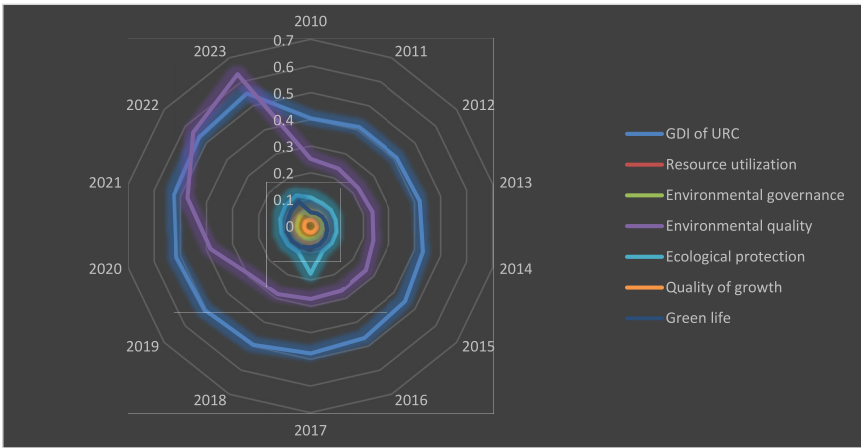


FIGURE 1
GD level using T-entropy weight.

TABLE 2 The mean value of first-level indicators in China based on the T-entropy weight.

Year	GDI of URC	Resource utilization	Environmental governance	Environmental quality	Ecological protection	Quality of growth	Green life
2010	0.404	0.057	0.040	0.253	0.105	0.024	0.052
2011	0.413	0.059	0.041	0.240	0.093	0.023	0.053
2012	0.413	0.059	0.042	0.229	0.095	0.023	0.056
2013	0.421	0.059	0.043	0.239	0.095	0.023	0.059
2014	0.433	0.061	0.044	0.242	0.098	0.024	0.064
2015	0.454	0.062	0.045	0.266	0.100	0.022	0.070
2016	0.467	0.063	0.047	0.269	0.102	0.023	0.077
2017	0.479	0.063	0.048	0.274	0.179	0.024	0.081
2018	0.496	0.064	0.050	0.284	0.107	0.024	0.087
2019	0.507	0.065	0.051	0.293	0.110	0.025	0.09
2020	0.517	0.066	0.053	0.386	0.113	0.025	0.092
2021	0.526	0.070	0.053	0.475	0.117	0.025	0.089
2022	0.535	0.061	0.055	0.565	0.121	0.026	0.094
2023	0.553	0.064	0.058	0.634	0.125	0.025	0.102

for 286 analyzed Chinese cities between 2010 and 2023. To facilitate the analysis of the overall characteristics of GDI for URC in China, we take the average of the GDI and the six first-level indicators. The relevant results are presented in Figure 1 and Table 2.

It can be clearly observed that, on the whole, the GD of URC in China presents a trend of steady improvement. Under the TOPSIS-Entropy Weight calculation, standardized GD of URC has steadily risen from 0.404 in 2010 to 0.553 in 2023, marking a total rise of 38.88% and an average annual increase rate of 2.63%. Among all six first-level indicators, environmental quality notably exerts the most significant influence on the GD of URC. Additionally, China's environmental quality demonstrates a yearly improvement trend.

TABLE 3 KMO and bartlett tests.

Inspection method	Test value	Inspection result
KMO test	0.609	Suitable for factor analysis
Bartlett's sphericity test (p-value)	0	Suitable for factor analysis

Green life also substantially impacts the GD of URC and similarly exhibits an annual improvement trend. Other first-level indicators have a positive impact on China's GD as a whole, and the trend is increasing year by year.

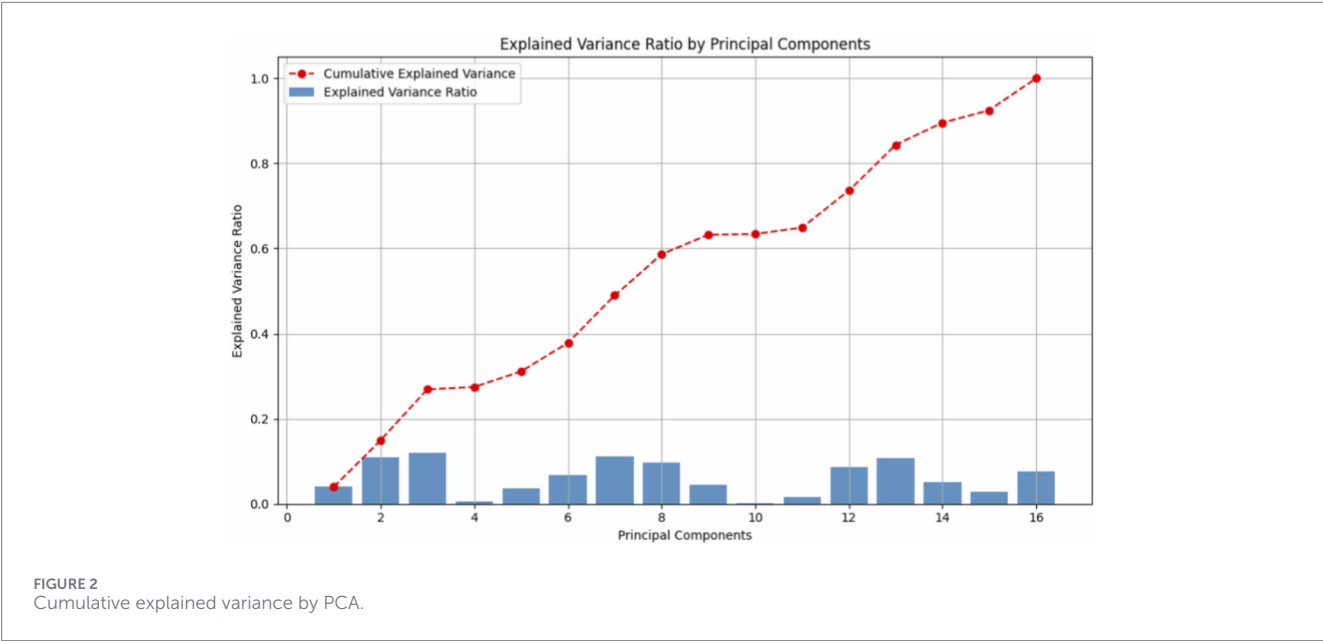


TABLE 4 Factor load matrix after rotation.

Explanation	Variable	Factor 1	Factor 2	Factor 3	Factor 4	Factor 7
Water consumption/10 ⁴ CNY of GDP	WC			0.6		
10 ⁴ CNY of GDP industrial waste emissions	WE				0.7786	
Average PM _{2.5} concentration and annual ratio	PM	0.5781				
Forest coverage	FC	0.5737				
Green coverage of built-up areas	GA		0.5683			
Change rate of the mean value of cultivated land	CL					0.5201
Road area increment (2005 is the fixed base)	RAI	0.5801				
Built-up area increment (based on 2005)	AI	0.652				
Per capita GDP growth rate	GR	0.5148				
Proportion of employees in the secondary industry	PE		0.6089			

TABLE 5 Clustering results of GD of URC in China in 2023.

Group	Number of cities	Grade	GD index range for urban and rural development
First category	29	Advantage	[0.674,0.817]
The second category	68	Second-best	[0.565,0.667]
The third category	120	Average	[0.484,0.574]
The fourth category	69	Weaker	[0.327,0.490]

Since the entropy weight algorithm aims to regulate all data to [0,1], the data values obtained are too small.

Nevertheless, the comprehensive level of GD in China’s URC, as assessed using the entropy weight method, remains relatively low. Based on 2023, it is only 55.3% of the highest level. There remains significant potential for enhancing the GD of URC in China, necessitating that governments at all levels implement appropriate policies to support and expedite this development.

4.2 Cluster analysis

The Ward method of system clustering is used to cluster the indices. The clustering quantity is established upon the fundamental principle where the sum of squared deviations for similar indices remains low, while the sum of squared deviations for evaluation indices across diverse categories is high, and the total sum of squares of deviations for all categories should be minimized. The application of the cluster analysis method in the study of GD has made considerable achievements (Razzaq et al., 2023). Cluster analysis is

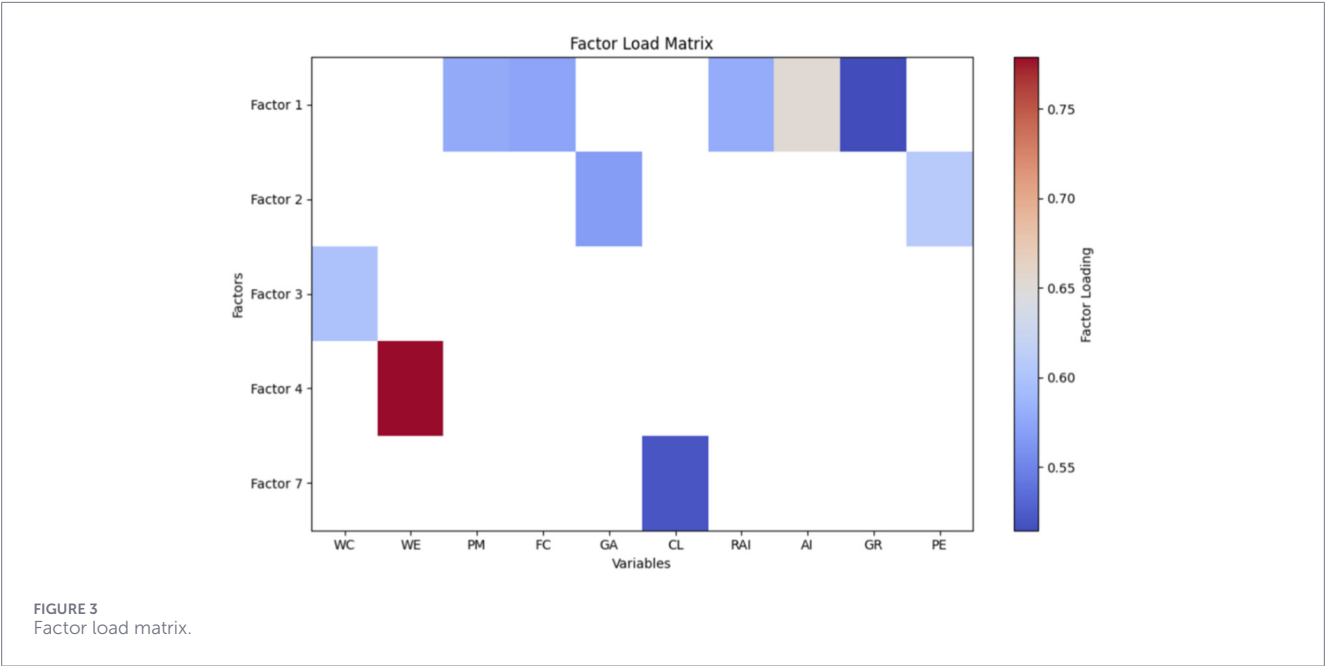


TABLE 6 The mean value of GD in URC and the mean value of the first-level indicators.

Group	GDURC	RU	EG	EQ	EP	QG	GL
1	0.735	0.071	0.063	0.235	0.213	0.024	0.129
2	0.612	0.069	0.061	0.2	0.15	0.028	0.104
3	0.529	0.066	0.056	0.175	0.11	0.027	0.095
4	0.448	0.051	0.054	0.132	0.089	0.023	0.099

GDURC: GD, in urban and rural areas; RU, resource utilization; E.G, environmental governance; EQ, environmental quality; EP, ecological protection; QG, quality of growth; GL, green life.

often combined with the PCA method. On the one hand, PCA can be used to achieve dimension reduction. On the other hand, the scientific research method can be guaranteed. Of course, it should be noted that technical problems involving the cluster analysis method and PCA method are beyond the scope of this paper. In this paper, we further deploy the PCA to reduce the dimension of variables affecting the GD of URC, and then the cluster analysis method will be used to classify the GD of URC in China.

1. Conducting PCA

Before performing factor analysis, we conduct Bartlett’s and Kaiser–Meyer–Olkin (KMO) tests for the fitness of the index dataset for factor analysis. The higher fitness is demonstrated when the KMO value is closer to 1, and the p-value by calibrating Bartlett’s test is less than 0.05 significance level. In this paper, 286 cities in China in 2023 are taken as examples. The outputs are provided in Table 3 that the KMO test value is 0.609 > 0.5, and the Bartlett test $P = 0.000 < 0.05$. It can be demonstrated that the fitness for factor analysis of the existing data and the validity of the outcomes.

2. Determination of the common factor and load value

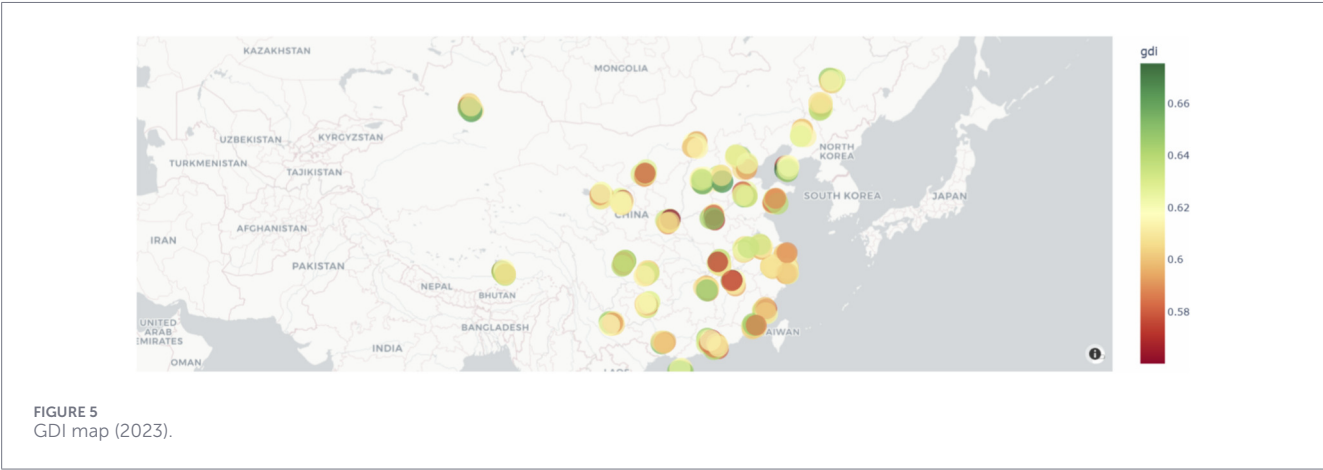
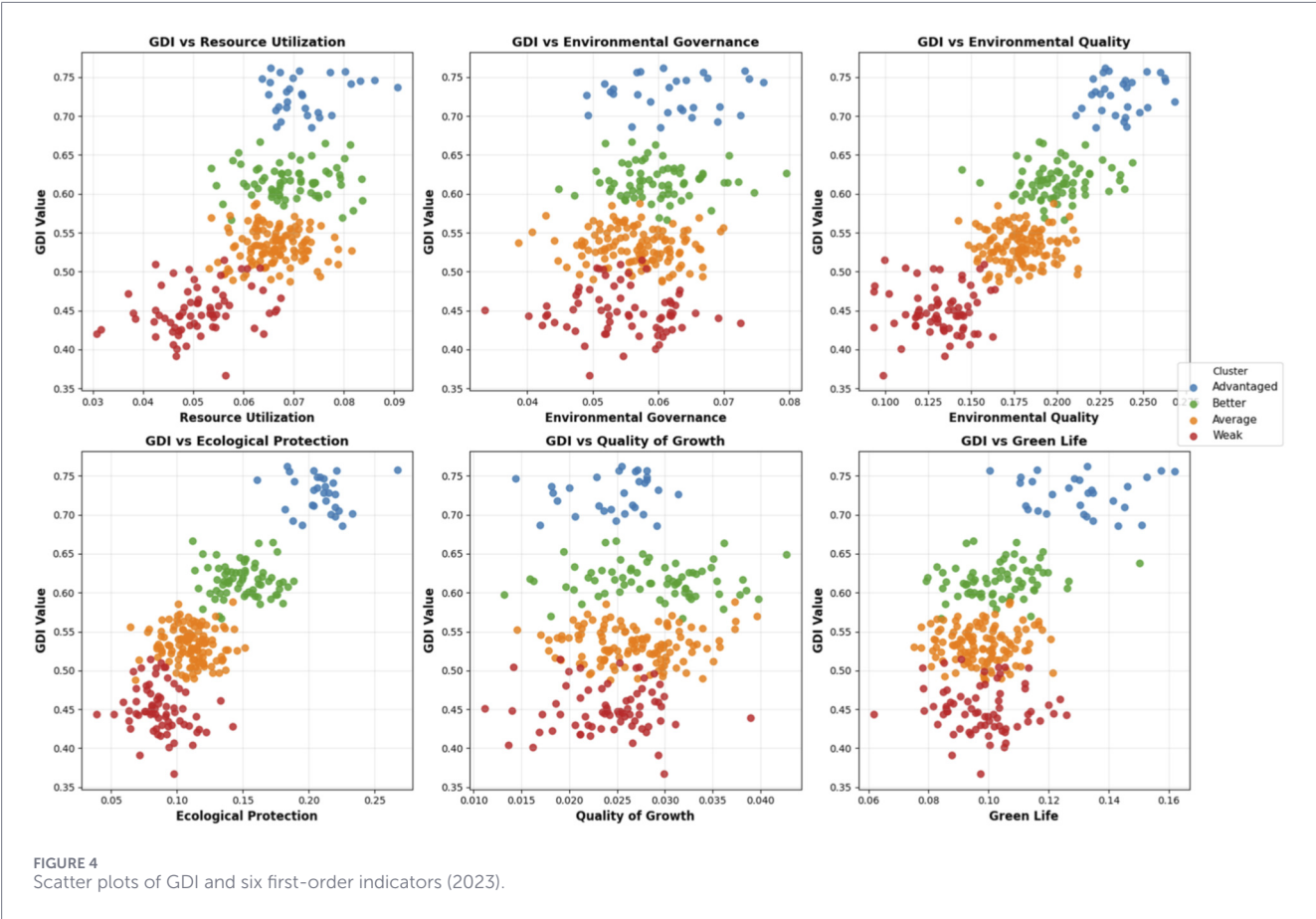
According to the calculation results of PCA, there were 16 factors with a common factor eigenvalue greater than 1, indicating that 16 factors could fully explain the variables represented by the original 42 indicators, and the total explanatory variance was 63.37%. Meanwhile, Figure 2 also shows that the first 16 factors have sufficient explanatory ability. Therefore, taking 16 factors as public factors can accurately reflect the GD level of URC in 286 cities in 2023.

In order to better select variables with strong explanatory power, rotation analysis was carried out on 16 common factors to obtain the load value of each variable for each factor. The load value whose absolute value was greater than 0.5 was reserved, and the factor load matrix after rotation was synthesized. Relevant results are shown in Table 4. According to relevant calculation results, there are only 5 factors with strong explanatory power, and there are 10 corresponding variables.

3. Cluster analysis

According to the 10 variables of factor analysis load mentioned above, the GD level of URC of 286 cities in China is systematically clustered. The relevant results are depicted in Table 5 and Figure 3. The findings reveal that according to the GD level of URC of 286 cities in China, they can be divided into 29 advantageous cities, 68 second-best cities, 120 average cities, and 69 weak cities.

Category I URC GD index interval [0.674,0.817]. As can be seen from Table 6, Category I cities are superior to other cities in resource utilization, environmental regulation, air quality, ecological preservation, green life, and other aspects of all first-level indicators, especially in environmental protection. Therefore, in the overall city sample, it is classified as a URC GD advantage city. The cities in the Advantaged group have natural endowment, favorable environmental governance, and high green infrastructure development. One of the cities in the Advantaged group is Lhasa,



which is known for having a high forest cover, low levels of industrial pollution, and a high reliance on natural sources of energy, which is common in Tibet. This ensures high levels of air and water quality as well as high cultivated land size. This is also the case with some cities located in the Pearl River Delta region.

Based on the same analysis, the GD index interval of the second type of city construction [0.565,0.667] is stronger than that of other analog cities in terms of growth quality among all first-level indicators, but weaker than that of the first type of city in other

aspects. Therefore, in the overall city sample, it is classified as the second-best city for GD of URC. According to the above analysis, the GD level of URC of dominant cities and second-best cities is higher than the average level of China (0.553). The total number of cities is 97, accounting for 33.92% of all the cities studied, indicating that the GD of URC of most cities in China needs to be improved. These cities generally experience rapid economic transformation in the fields of service and high-tech industries; however, they still experience the transitional pressure from traditional manufacturing. The cities represented by Shenzhen and Guangzhou are typical

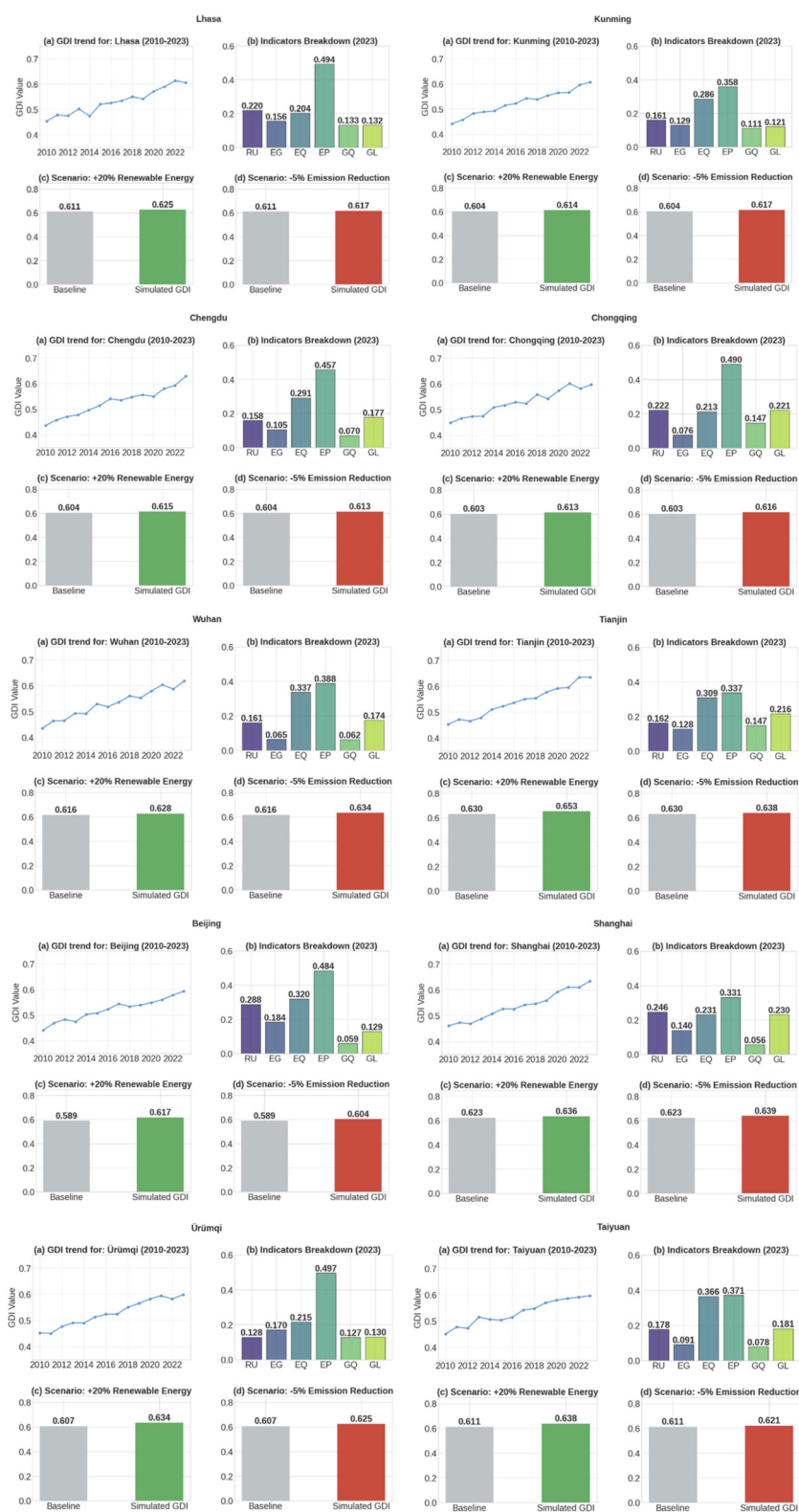


FIGURE 6
GDI analysis and simulation for typical cities.

TABLE 7 Elasticity coefficients used in the policy simulator.

First-level indicator	Action 0: +20% renewable energy share (5-year)	Action 1: 5% emissions (5-year)	R ₂ of regression
RU	+0.087	+0.021	0.74
EG	+0.041	+0.068	0.81
EQ	+0.112	+0.079	0.89
EP	+0.098	+0.033	0.77
QG	+0.054	+0.026	0.69
GL	+0.067	+0.041	0.72

examples that embody the above characteristics with high growth rates in the cities' *per capita* GDP and urbanization rates, as well as continuous improvements in the cities' green life indices, while still facing the task of further improving the work on ecological protection.

The range of the third category of GDI of urban construction [0.484,0.574] is above and below the average level of GD of URC in China, and all indicators are at the middle level among the average of all first-level indicators. There are 120 cities, accounting for 41.96% of all cities studied. Therefore, in the overall city sample, it is classified as URC, GD, second-best city, and average city. Many cities in this cluster are located in central and western inland regions, where mixed industrial structures persist alongside moderate urbanization. For instance, cities like Wuhan and Chengdu typically show average environmental quality and ecological protection, with ongoing land conversion and industrial activities constraining faster green progress despite steady economic growth.

In the fourth category of URC GDI interval [0.327,0.490], in the average of all first-level indicators, all indicators are in other cities, especially in environmental quality and ecological protection, where there is a big gap with other cities. Therefore, it is classified as a city with weak GD in the URC in the overall city sample. Although the four analogies of urban GD level based on cluster analysis overlap a little, they still show the differences between each analogy. Figure 3 above illustrates the relevant differences more vividly. The cities generally experience pressure from intensive resource extraction, high dependence on the industrial economy, and the destruction of natural habitats. The cities represented by the resource-based cities in the western and central areas, such as Urumqi, as well as the cities represented by the industrial cities in Shanxi Province and Inner Mongolia, are typical examples that embody the above characteristics with the deterioration of air quality, the reduction in the coverage of forests and farmlands, as well as the difficulty in the development of green infrastructure. The pronounced gaps in environmental quality and ecological protection between the Advantaged and Weak groups (approximately 0.103 and 0.124, respectively) underscore that these two dimensions serve as the primary drivers of differentiation across Chinese cities.

Figure 4 shows scatter plots of the association between the overall GDI and each of the six first-level indicators. The panel for EQ reflects the substantial influence of the rate of change of average PM_{2.5} concentration, industrial sulfur dioxide emissions, and industrial wastewater discharge. Cities in the Advantaged cluster consistently occupy the upper-right region of this panel, indicating effective control of fine particulate matter and industrial pollutants. Similarly, the Ecological Protection panel demonstrates an even clearer positive relationship with GDI, driven by secondary indices including forest coverage, green space coverage in built-up areas, road area increment (with 2005 as the fixed base), change rate of cultivated land, and built-up area increment. Advantaged cities such as Lhasa exhibit high values in both forest coverage and green space coverage, placing them in the high-GDI, high-EP quadrant, whereas Weak-cluster cities (certain resource-based cities in western China) cluster in the lower-left area due to notable losses in cultivated land and limited green space expansion.

In contrast, the scatter plots for Quality of Growth and Resource Utilization display more dispersed distributions. The Quality of Growth panel, which incorporates secondary indices such as *per capita* GDP growth rate, urbanization rate, and the proportion of the tertiary sector's value-added to GDP, shows moderate correlation with overall GDI, suggesting that economic expansion alone does not guarantee strong GD outcomes. Likewise, the Resource Utilization panel, linked to secondary measures including carbon dioxide emissions per GDP unit, water consumption per GDP unit, and incremental power usage per GDP unit, reveals weaker discriminatory power across clusters.

4.3 Policy simulations and implications

This section evaluates the performance of the DQN model in optimizing GDI policies for Chinese prefecture-level cities. The GDI map can be found in Figure 5. Additionally, Figure 6 reveals a stratified landscape of urban GD across the four designated clusters for ten typical cities. All four groups show steady improvement in GDI over the 13 years. However, they differ in their levels, growth rates, and support policies. For instance, in cluster 1, where cities such as Lhasa and Kunming are found, there exist high levels of performance in EP, where the level is 0.494 and 0.358, respectively. There also exist high levels of performance in terms of EQ, where the level is 0.204 and 0.286, respectively, because of the high level of forest cover, low level of industrial pollution, and renewable energy-based economy in this region. The balance of tourism and biodiversity in the Dianchi Lake region is an example of a high level of performance in this region. In contrast, in cluster 2, where cities such as Shanghai, Chengdu, and Chongqing are found, there exist high levels of performance in terms of RU, where the level is 0.246. There are strong levels of performance in terms of GL, with a level of 0.230. This is due to the service-based economy, smart grids, and high infrastructure penetration in the cities. The cities in the region have also been successful in the transition to electric vehicles and green supply chains, as noted by Ngoc et al. (2024).

Cluster 3 (Wuhan, Tianjin, Beijing) shows the strong EQ (0.337) and solid GL (0.221). Beijing performs well in public amenities. However, high population density and legacy industries (steel and logistics in Wuhan and Tianjin) limit faster progress in resource utilization. Despite significant, E.G., expenditures, driven by central

TABLE 8 Sensitivity analysis of the reward function coefficients.

Scenario	α	β	Avg. Cumulative reward (renewable energy +20%)	Avg. Cumulative reward (emission reduction -5%)	Preferred policy ($\geq 94\%$)
Final calibrated	3.155	0.362	0.181	0.089	Renewable energy +20%
α reduced by 25%	2.366	0.362	0.159	0.092	Renewable energy +20%
α increased by 25%	3.944	0.362	0.198	0.087	Renewable energy +20%
β increased by 50% (high cost)	3.155	0.543	0.147	0.104	Renewable energy +20%
β reduced by 50% (low cost)	3.155	0.181	0.204	0.093	Renewable energy +20%

TABLE 9 Implications of the four-city typology.

Cluster	No. of cities	Recommended policy
Advantaged	29	National demonstration zones; pilot carbon-neutral districts, zero-waste cities, and green finance innovation
Second-best	68	Fast-follower strategy; scale up renewable portfolio and green industry clustering within 3–5 years
Average	120	Core battlefield; enforce mandatory targets for EQ and EP improvement; tie fiscal transfers to annual GDI gains
Weak	69	Phase out backward capacity, relocate polluting firms, and import clean energy

government mandates, these cities are burdened by high population density and industrial legacies. Beijing's GDI is suppressed by the energetic demands of urban heating and massive traffic volumes. Similarly, Wuhan and Tianjin act as important logistics and steel hubs, where the elasticity of green transition faces challenges concerning the complexity of local labor markets. Lastly, concerning cluster 4: Taiyuan, Urumqi, the RU is very low at 0.178, whereas the EP is relatively higher at 0.484. Taiyuan has a high dependence on industries related to coal-based activities in Shanxi Province. The environmental development path of Taiyuan faces challenges related to the high carbon-based composition of its energy mix, in which there is a need for significant technological interventions such as Carbon Capture, Utilization, and Storage (CCUS). Urumqi faces challenges related to fossil fuels and arid environments that expose the city to desertification. At this point, there are inefficiencies in the pattern of resource utilization. For these cities, the GDI development path from 2010 to 2023 indicates more volatility, which means a high sensitivity to variations in the macroeconomic environment and to variations in energy prices.

The DQN simulations show that increasing renewable energy by 20% generally produces stronger effects than reducing emissions by 5%, especially in the Weak and Average groups. The largest gain from the renewable-energy scenario appears in the Weak group (+0.0266). Table 7 reveals that a 20% increase in renewable energy share generates 1.4–5.2 times larger improvements in

Environmental Quality ($\Delta EQ = +0.112$) and Ecological Protection ($\Delta EP = +0.098$) than a 5% emission-reduction policy. This explains the consistently higher rewards of the renewable-energy action observed across all scenarios in Table 8. Even when the cost penalty β is increased by 50% or the EQ bonus α is reduced by 25%, the renewable-energy pathway remains strictly dominant (selected in $\geq 94\%$ of episodes).

Based on the clustering analysis, there is still a large difference in urban and rural GD among cities in China. There are still a few cities with development advantages and sub-optimal cities. GD cities in urban and rural development make up the main part, while some cities are underdeveloped. Environmental quality and ecological protection are the main factors of significant differences between cities. Hence, appropriate measures should be implemented to enhance environmental quality and advance ecological protection, as in Table 9.

From policy simulation, we can observe that low reliance on fossil fuels results in a substantial GDI gain. This is in line with the renewable energy policy objectives set in the 14th Five-Year Plan in China, as well as the recommendations from the previous literature on the subject (Xiao et al., 2022), focusing on clean energy for the achievement of carbon neutrality by 2060. In the case of Beijing, it is also clear that the policy of using renewable energy will lead to an increase in GDI. This is due to the fact that, despite the complexity of the industrial structure, there is still a benefit from the use of clean forms of energy. The effectiveness of the policy is also attributed to the direct impact on energy consumption patterns, which is a universal contributor to the GDI in all cities. The policy is therefore highly effective in low-as well as high-emission cities, as it bypasses the complexity in the emission from industries. The effectiveness of the policy is also limited by the complexity in the retrofitting process, which requires significant investment for the achievement of the desired outcomes. The effectiveness of emission reduction policies is also subject to the level of pollution in the respective cities. The policy simulation plot shows that the renewable energy policy is the best policy in terms of emission reduction compared to the emission reduction policy. This is because renewable energy deals with the root cause of the degradation of the environment, i.e., the use of fossil fuels. The renewable energy policy is also more effective as the action includes the implementation of forest-related pathways such as biomass power, biomass heating, and agro-forestry-photovoltaic hybrids (Slimene et al., 2026; Ta et al., 2025b; Bui-Duy et al., 2023), resulting in an increased GDI in three out of the four clusters as the

policy increases the three high-weight dimensions in the index, i.e., EP, EQ, and RU.

The provinces with similar natural and socioeconomic conditions, such as Guangdong, Guizhou, Fujian, Jiangxi, Hunan, and the southern part of Shaanxi, have more than 28 million hectares of fast-growing plantations and protective forests around the reservoirs, which can be immediately converted into “PV forestry” or “wind forestry” without the need for additional land clearance, potentially adding 35–45 GW clean power by 2030 while increasing the forest cover by at least 0.3 percent nationally. This policy recommendation is in line with the recommendations from the previous literature on the subject as presented in Xu et al. (2025). Priority should be given to transferring the Guangxi model (large-scale bifacial PV over eucalyptus plantations combined with green hydrogen electrolysis) to the remaining five provinces of the Pearl River Basin and the upper-middle Yangtze region, where annual solar irradiation exceeds 1,300 kWh/m² and existing pulpwood cycles are 5–7 years.

5 Concluding remarks

Incorporating remote sensing and statistical data, this study has addressed the essential aspects by building a series of index systems, quantifying the GD of URC across cities using a unified standard, and analyzing the characteristics of the GD of URC in China. To examine the GD level of URC in China, this paper analyzes 286 over-prefecture cities between 2010 and 2023. It constructs the GDI system of URC in China based on six first-level and 42 second-level connotations and employs the entropy weight method for measurement. Finally, the GD level of URC in China in 2023 was classified by the clustering method. The results show that: (1) On the whole, the GD of URC in China presents a trend of steady improvement. Under the TOPSIS-Entropy Weight calculation, the standardized GD of URC in China increases from 0.404 in 2010 to 0.553 in 2023, equivalent to approximately 38.88% relative to the 2010 value. (2) Environmental quality has the greatest impact on the GD of URC, and China's environmental quality demonstrates a consistent upward trend year by year, indicating improvement. Green life notably influences the GD of URC and exhibits a similar positive trend of enhancement over time. (3) There is still a large difference in urban and rural GD among cities in China. Among all the cities studied, cities with advantages and sub-optimal development account for 33.92%, cities with average GD account for 41.96%, and there are still some cities with weak development. (4) The main factor of the GD difference between URC is the difference in environmental quality and ecological protection. Hence, to advance GD across urban and rural regions universally, authorities at all administrative tiers should prioritize initiatives geared towards fostering high-quality environmental advancement. URC GD average and weak cities, but also should increase the intensity of environmental protection. Analysis of the policy simulation plot from the DQN model reveals that increasing renewable energy by 20% performs better than reducing emissions by 5%. This research also recommends prioritizing renewable energy for less industrialized cities and emission reductions for urban centers. This aligns with findings from previous studies (Hsu et al., 2025; Li X. et al., 2024; Ran et al., 2024; Dinh-Quoc et al., 2025). These findings support China's carbon neutrality

goals by identifying scalable, high-impact strategies for urban sustainability.

Data availability statement

The original contributions presented in the study are included in the article/Supplementary Material, further inquiries can be directed to the corresponding author.

Author contributions

XH: Conceptualization, Supervision, Formal Analysis, Methodology, Writing – original draft, Data curation, Software, Visualization, Investigation, Resources, Validation, Writing – review and editing.

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Author XH was employed by Jiangsu Rongzer Information Technology Corp.

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The author(s) declared that generative AI was not used in the creation of this manuscript.

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Supplementary material

The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/feart.2026.1802621/full#supplementary-material>

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